Databricks Trainer

# Question bank

with answers and explanations

205 multiple-choice questions across three tracks.

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## How to use this document

Each question shows its options, the correct answer (highlighted in green and repeated below the question) and a short explanation of why it is right. To test yourself, cover everything below the options and compare with your pick. A compact answer key is at the end.

Option order is shuffled with a fixed seed, so the correct answer is not always the same letter, and a given letter is stable between runs. The interactive trainer shuffles options differently on each attempt.

Exam and architecture content was written from general knowledge of Databricks and Azure Databricks. Check details against the current documentation, because features and names change often.

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# Data Engineer Associate exam

## Databricks Intelligence Platform

Lakehouse architecture, workspace, compute choices, notebooks and Git folders.

**QUESTION 1**

**Where is the table data of a Delta table physically stored when using classic compute?**

A. In the driver node's local disk

B. In the Databricks control plane

C. In the Unity Catalog metastore database

**D. ✓ In cloud object storage in the customer's account**

> **Answer: D** Table data (Parquet files plus the _delta_log) lives in cloud object storage. The control plane holds metadata, notebooks and job definitions, but not table data.

**QUESTION 2**

**A data engineer needs to run a production ETL job every night at the lowest cost. Which compute should they use?**

**A. ✓ A job cluster (or serverless jobs compute)**

B. A single-node all-purpose cluster with auto-termination disabled

C. A shared all-purpose cluster that stays on

D. A SQL warehouse

> **Answer: A** Job clusters are created for a run and terminated afterwards, and they cost fewer DBUs than all-purpose compute. All-purpose clusters are meant for interactive work.

**QUESTION 3**

**Which component of a Delta table records every transaction and enables time travel?**

**A. ✓ The _delta_log directory**

B. The Hive metastore

C. The Spark UI

D. The cluster event log

> **Answer: A** The _delta_log folder holds ordered JSON commits (and periodic Parquet checkpoints). Readers rebuild the table state from it, which is also what makes time travel possible.

**QUESTION 4**

**What is the effect of `%run ./setup` in a notebook cell?**

A. It runs ./setup on a new cluster and returns its exit value

B. It imports ./setup as a Python module

**C. ✓ It runs ./setup inline, so functions and variables it defines become available in the current notebook**

D. It schedules ./setup as a separate job

> **Answer: C** %run executes the other notebook in the same context. dbutils.notebook.run() is the one that starts a separate run and returns a value.

QUESTION 5

**Which feature reduces cluster start and autoscaling time by keeping idle, ready-to-use instances?**

A. Spot instances

B. Auto termination

**C. ✓ Cluster pools**

D. Photon

> **Answer: C** Pools keep a set of idle instances, so clusters attached to the pool can acquire nodes quickly.

QUESTION 6

**A team wants to version-control notebooks in GitHub and create branches from the Databricks UI. What should they use?**

**A. ✓ Git folders (formerly Repos)**

B. Cluster init scripts

C. DBFS mounts

D. Notebook revision history only

> **Answer: A** Git folders clone a remote repo into the workspace and support branching, commit, push and pull from the UI.

QUESTION 7

**What is Photon?**

A. A data catalog for lineage

B. A library for ML model serving

**C. ✓ A vectorized query engine that speeds up SQL and DataFrame workloads**

D. A streaming source connector

> **Answer: C** Photon is a native vectorized engine compatible with Spark APIs. It speeds up queries with no code changes.

QUESTION 8

**A notebook needs a parameter `run_date` that a job can set. How should the notebook read it?**

```
# inside the notebook
```

A. sys.argv[1]

**B. ✓ dbutils.widgets.get('run_date')**

C. spark.conf.get('run_date')

D. os.environ['run_date']

> **Answer: B** Job and task parameters reach notebooks as widgets, which you read with dbutils.widgets.get().

QUESTION 9

**Which statement about restarting an all-purpose cluster is true?**

A. It deletes the cluster configuration

**B. ✓ It clears in-memory state, such as variables and cache, for attached notebooks**

C. It deletes tables created on that cluster

D. It keeps cached data and Python variables

**Answer: B** A restart clears the notebook state on the cluster. Tables in storage are unaffected, and the configuration is kept.

QUESTION 10

**Which compute type is purpose-built for SQL queries and BI dashboards?**

A. All-purpose cluster

B. Job cluster

**C. ✓ SQL warehouse**

D. Single-node cluster

**Answer: C** SQL warehouses run Databricks SQL with Photon, scale for concurrency, and are what BI tools connect to.

QUESTION 11

**A library was installed with `%pip install` in a notebook. What happens after the cluster restarts?**

A. The cluster refuses to restart

B. It stays installed for everyone

C. It's promoted to a cluster library

**D. ✓ It has to be installed again: notebook-scoped libraries last only for that notebook's session**

**Answer: D** %pip installs into the notebook's Python environment, which is reset when the cluster restarts or the notebook detaches.

QUESTION 12

**Which magic command runs shell commands on the driver node?**

A. %run

B. %fs

**C. ✓ %sh**

D. %md

**Answer: C** %sh runs a shell on the driver. %fs is a shortcut for dbutils.fs, %run runs another notebook inline, and %md renders Markdown.

QUESTION 13

**A cluster has auto termination set to 30 minutes. What does that mean?**

A. It restarts every 30 minutes

B. Jobs time out after 30 minutes

**C. ✓ It terminates after 30 minutes with no activity**

D. It shuts down 30 minutes after it starts

**Answer: C** Auto termination stops an idle cluster, so you don't pay for compute nobody is using.

QUESTION 14

**A notebook must call another notebook, pass it parameters and use the value it returns. Which option fits?**

**A. ✓ dbutils.notebook.run('./child', 600, {'x': '1'})**

B. import child

C. spark.sql('RUN child')

D. %run ./child

**Answer: A** dbutils.notebook.run starts the child as its own run with parameters and a timeout, and returns what the child passes to dbutils.notebook.exit().

QUESTION 15

**In which file format does Delta Lake store table data?**

**A. ✓ Parquet**

B. Avro

C. ORC

D. CSV

**Answer: A** Delta tables are Parquet data files plus the _delta_log transaction log.

QUESTION 16

**Which statement about serverless compute is correct?**

A. It runs in your own VNet

**B. ✓ Databricks manages the compute, which starts in seconds without cluster configuration**

C. You choose the VM types and Spark configuration

D. It's only available for Python

**Answer: B** Serverless compute is provisioned and scaled by Databricks in its own compute plane, so there's nothing to size or configure.

QUESTION 17

**Which Databricks Runtime should a production workload use for long-term stability?**

A. The newest beta runtime

**B. ✓ An LTS (long-term support) runtime**

C. The runtime of the developer's laptop

D. Any ML runtime

**Answer: B** LTS runtimes are supported for longer and get stability fixes, which suits production.

QUESTION 18

**What is a key advantage of the lakehouse over a separate data lake plus data warehouse?**

A. It removes the need for any storage

**B. ✓ One copy of data in an open format serves BI and ML, with ACID transactions and governance**

C. It only supports streaming

D. Data must be copied into a proprietary format

**Answer: B** The lakehouse combines open, low-cost storage with warehouse features (ACID, schema enforcement, governance) on a single copy of the data.

## Development & Ingestion

Reading files, CTAS, COPY INTO, Auto Loader and Lakeflow Connect.

**QUESTION 19**

**A job runs `COPY INTO bronze.orders FROM '/landing/orders' FILEFORMAT = JSON` every hour. What happens to files that were already loaded in an earlier run?**

A. The command fails

B. They are loaded again, which creates duplicates

C. They are deleted from the source directory

**D. ✓ They are skipped, because COPY INTO is idempotent**

> **Answer: D** COPY INTO tracks which files it has loaded and skips them, so re-running it is safe.

**QUESTION 20**

**Millions of JSON files arrive continuously in cloud storage. Which is the most scalable way to ingest only new files?**

**A. ✓ Auto Loader (cloudFiles)**

B. CREATE TABLE AS SELECT on a schedule

C. INSERT INTO ... SELECT * FROM json.`path`

D. Manually listing files with dbutils.fs.ls

> **Answer: A** Auto Loader is built for scalable incremental file discovery, with file notification mode and schema inference and evolution.

**QUESTION 21**

**In Auto Loader, what is the purpose of `cloudFiles.schemaLocation`?**

A. It is where malformed records are written

**B. ✓ It stores the inferred schema and tracks schema evolution**

C. It sets the location of the raw source files

D. It sets the location of the output Delta table

> **Answer: B** Auto Loader saves the inferred schema at schemaLocation and updates it as new columns appear.

**QUESTION 22**

**Auto Loader encounters fields that don't match the expected schema. Where does that data end up by default?**

A. The stream always fails immediately

B. It is silently dropped

**C. ✓ In the _rescued_data column**

D. In a separate quarantine table

> **Answer: C** The rescued data column captures values that couldn't be parsed into the schema, so no data is lost.

QUESTION 23

**Which statement about CREATE TABLE AS SELECT (CTAS) is correct?**

A. It loads data incrementally on each run

B. It only works on Parquet files

**C. ✓ The schema is inferred from the query results**

D. You must declare column types explicitly

> **Answer: C** CTAS infers the schema from the query and doesn't accept manual column type declarations. It's a one-time load.

QUESTION 24

**How do you query a directory of CSV files that have a header row, without creating a table?**

A. LOAD CSV '/path'

B. SELECT * FROM text.`/path`

C. SELECT * FROM csv.`/path` WITH HEADER

**D. ✓ SELECT * FROM read_files('/path', format => 'csv', header => true)**

> **Answer: D** read_files() accepts reader options such as header and delimiter. Direct path queries like csv.`/path` can't pass options.

QUESTION 25

**What is the key difference between INSERT OVERWRITE and CREATE OR REPLACE TABLE ... AS SELECT?**

**A. ✓ INSERT OVERWRITE only replaces data and needs a matching schema. CREATE OR REPLACE can change the schema.**

B. INSERT OVERWRITE can change the table schema. CREATE OR REPLACE can't.

C. CREATE OR REPLACE appends data

D. They're identical

> **Answer: A** INSERT OVERWRITE replaces the data but keeps the existing schema. CREATE OR REPLACE TABLE replaces the whole table definition.

QUESTION 26

**An analyst wants to query a PostgreSQL database from Databricks without copying the data. Which feature should they use?**

A. Auto Loader

**B. ✓ Lakehouse Federation**

C. COPY INTO

D. Delta Sharing

> **Answer: B** Lakehouse Federation creates foreign catalogs that query external databases in place.

QUESTION 27

**Which query returns one row per line of the log files in a directory?**

A. SELECT * FROM json. `/logs/`

B. SELECT * FROM binaryFile. `/logs/`

**C. ✓ SELECT * FROM text. `/logs/`**

D. SELECT * FROM delta. `/logs/`

> **Answer: C** The text format returns a single 'value' column with one row per line. binaryFile returns one row per file.

QUESTION 28

**Which COPY INTO setting lets new columns in the source files be added to the target table?**

A. FORMAT_OPTIONS ('header' = 'true')

B. VALIDATE ALL

C. PATTERN = '*.json'

**D. ✓ COPY_OPTIONS ('mergeSchema' = 'true')**

> **Answer: D** mergeSchema in COPY_OPTIONS allows schema evolution when new columns appear.

QUESTION 29

**With Auto Loader's default schema evolution mode, what happens when a new column appears in incoming files?**

A. The new column replaces an old one

B. All existing data is rewritten

C. The column is silently ignored forever

**D. ✓ The stream stops with an error after adding the column to the schema, and picks it up when restarted**

> **Answer: D** The default addNewColumns mode updates the stored schema and fails the stream once. A restart (for example a job retry) continues with the new column.

QUESTION 30

**Which SQL table function reads files with options, and is used with STREAM in declarative pipelines?**

A. load_files

B. cloud_read

C. file_scan

**D. ✓ read_files**

> **Answer: D** read_files('path', format => 'json', ...) reads files in batch, and STREAM read_files(...) ingests them incrementally with Auto Loader.

QUESTION 31

**What does selecting `_metadata.file_path` return when reading files?**

**A. ✓ The path of the source file each row came from**

B. The checkpoint path

C. The notebook path

D. The table's storage location

**Answer: A** The hidden _metadata column exposes file-level information such as file_path, file_name and file_modification_time. That's useful in bronze tables.

QUESTION 32

**How long does a temporary view created with `CREATE TEMP VIEW` last?**

A. Forever

**B. ✓ Until the Spark session ends (for example, the notebook is detached or the cluster restarts)**

C. Until the next VACUUM

D. 7 days

**Answer: B** Temporary views belong to the session and aren't stored in the catalog.

QUESTION 33

**How do you query a global temporary view named `daily`?**

A. SELECT * FROM main.daily

B. SELECT * FROM daily

C. SELECT * FROM temp.daily

**D. ✓ SELECT * FROM global_temp.daily**

**Answer: D** Global temp views live in the special global_temp schema and are shared by notebooks on the same cluster.

QUESTION 34

**Which Databricks feature provides managed, incremental connectors for sources like Salesforce and SQL Server?**

A. Lakehouse Federation

B. Delta Sharing

C. COPY INTO

**D. ✓ Lakeflow Connect**

**Answer: D** Lakeflow Connect ingests from SaaS apps and databases into Delta tables. Federation queries sources in place without ingesting.

QUESTION 35

**Two Auto Loader streams write to two different tables. Can they share one checkpoint location?**

**A. ✓ No, each stream needs its own checkpoint location**

B. Only on serverless

C. Yes, checkpoints are shared by default

D. Only if they read the same files

**Answer: A** A checkpoint records one stream's progress and state. Sharing it corrupts both streams.

QUESTION 36

**Which is true about `INSERT INTO t SELECT ...` run twice with the same source?**

A. It fails because the rows exist

**B. ✓ The rows are appended twice, creating duplicates**

C. The second run is skipped automatically

D. It overwrites the table

> **Answer: B** INSERT INTO always appends. Use COPY INTO, Auto Loader or MERGE for idempotent loads.

QUESTION 37

**Which query reads JSON files directly without creating a table?**

**A. ✓ SELECT * FROM json.`/Volumes/main/raw/events/`**

B. SELECT * FROM TABLE('/Volumes/main/raw/events/')

C. LOAD JSON '/Volumes/main/raw/events/'

D. SELECT * FROM delta.json

> **Answer: A** Prefixing a path with a format (json., parquet., csv., text.) and backticks queries the files directly.

## Data Processing & Transformations

Spark SQL and PySpark transformations, Delta Lake operations and table maintenance.

QUESTION 38

**A bronze table has several rows per event_id. Which approach keeps exactly the latest row per event_id?**

A. GROUP BY event_id with no aggregates

B. SELECT DISTINCT * FROM bronze

**C. ✓ ROW_NUMBER() OVER (PARTITION BY event_id ORDER BY ingest_ts DESC) = 1**

D. DELETE FROM bronze WHERE event_id IS NULL

> **Answer: C** DISTINCT only removes rows identical in every column. A window function ranks the rows within each key so you can keep the first.

QUESTION 39

**Table t has 10 rows. The column email is NULL in 3 of them. What does `SELECT count(*), count(email) FROM t` return?**

A. 10, 10

B. 7, 7

C. 7, 10

**D. ✓ 10, 7**

> **Answer: D** count(*) counts all rows. count(column) skips NULLs.

QUESTION 40

**Which MERGE clause inserts only records that aren't already in the target, ignoring duplicates?**

A. WHEN MATCHED THEN UPDATE SET *

B. WHEN MATCHED THEN DELETE

**C. ✓ WHEN NOT MATCHED THEN INSERT ***

D. WHEN NOT MATCHED BY SOURCE THEN DELETE

> **Answer: C** An insert-only merge with just WHEN NOT MATCHED THEN INSERT * is the standard way to avoid inserting duplicates.

QUESTION 41

**After someone ran a bad UPDATE, how can you return a Delta table to version 5?**

**A. ✓ RESTORE TABLE t TO VERSION AS OF 5**

B. ROLLBACK TABLE t TO 5

C. VACUUM t RETAIN 5 VERSIONS

D. UNDO TABLE t

> **Answer: A** RESTORE TABLE ... TO VERSION AS OF (or TIMESTAMP AS OF) restores an earlier state, and is itself recorded as a new version.

QUESTION 42

**A user can no longer time travel to a 30-day-old version of a table. What is the most likely cause?**

**A. ✓ VACUUM removed the old data files**

B. The cluster was restarted

C. Z-ORDER was applied

D. OPTIMIZE was run

> **Answer: A** VACUUM deletes unreferenced files older than the retention period (7 days by default). Versions that relied on those files can no longer be read.

QUESTION 43

**Queries on a Delta table are slow because streaming writes produced many small files. Which command helps the most?**

A. REFRESH TABLE

**B. ✓ OPTIMIZE**

C. DESCRIBE HISTORY

D. VACUUM

> **Answer: B** OPTIMIZE compacts small files into larger ones. ZORDER BY or liquid clustering can also improve data skipping.

QUESTION 44

**Which higher-order function returns a new array with each element multiplied by 2?**

```
SELECT ____(nums, x -> x * 2) FROM t
```

**A. ✓ transform**

B. filter

C. exists

D. explode

> **Answer: A** transform applies a lambda to each element. filter keeps the elements that match a condition, and exists returns a boolean.

QUESTION 45

**In PySpark, which of these triggers execution of the query plan?**

A. .withColumn()

B. .groupBy()

**C. ✓ .count()**

D. .filter()

> **Answer: C** Transformations are lazy. Actions such as count, collect, show and write run the plan.

QUESTION 46

**A column `raw` holds JSON strings. Which Databricks SQL syntax extracts the nested field customer.id?**

A. raw.customer.id

B. json(raw)['customer']['id']

C. raw->customer->id

**D. ✓ raw:customer.id**

> **Answer: D** The colon path syntax (raw:customer.id) queries JSON strings directly. Dot syntax is for struct columns.

QUESTION 47

**Which command shows a Delta table's version history with the operation and user for each version?**

A. SHOW VERSIONS t

B. DESCRIBE DETAIL t

**C. ✓ DESCRIBE HISTORY t**

D. SHOW TBLPROPERTIES t

> **Answer: C** DESCRIBE HISTORY lists versions with timestamp, operation, user and metrics. DESCRIBE DETAIL shows location, size and file count.

QUESTION 48

**Which clause filters groups based on an aggregate, such as keeping only customers with more than 5 orders?**

A. WHERE

**B. ✓ HAVING**

C. QUALIFY

D. FILTER BY

> **Answer: B** WHERE filters rows before grouping, HAVING filters groups after aggregation, and QUALIFY filters on window function results.

QUESTION 49

**What does `a LEFT SEMI JOIN b ON a.id = b.id` return?**

**A. ✓ Rows of a that have a match in b, with only a's columns and no duplication**

B. All columns from a and b

C. Every combination of rows

D. Rows of a with no match in b

> **Answer: A** A semi join is an existence check. LEFT ANTI returns the rows without a match.

QUESTION 50

**What does `explode(items)` do to a row whose items array has 3 elements?**

A. Raises an error

**B. ✓ Returns 3 rows, one per element**

C. Returns 3 columns

D. Returns 1 row with a 3-element array

> **Answer: B** explode produces one output row per array element, repeating the other columns.

QUESTION 51

**Which higher-order function returns true if any element of an array matches a condition?**

**A. ✓ exists**

B. aggregate

C. filter

D. transform

> **Answer: A** exists(arr, x -> cond) returns a boolean. filter returns the matching elements, and transform maps every element.

QUESTION 52

**Which columns are the best candidates for `OPTIMIZE ... ZORDER BY`?**

A. Columns that are never queried

B. The partition columns

C. Low-cardinality columns such as country

**D. ✓ High-cardinality columns that are often used in filters, such as customer_id**

> **Answer: D** Z-ordering co-locates similar values so data skipping can prune files for selective filters on high-cardinality columns.

QUESTION 53

**What advantage does liquid clustering ( `CLUSTER BY` ) have over partitioning?**

A. It disables time travel

B. It only works on CSV

C. It needs no storage

**D. ✓ Clustering keys can be changed without rewriting existing data, and small-file and skew problems are avoided**

**Answer: D** Liquid clustering is the recommended layout for new Delta tables and replaces both partitioning and Z-order.

QUESTION 54

**What is VACUUM's default retention period?**

A. 24 hours

B. 30 days

C. No retention: it deletes all old files

**D. ✓ 7 days (168 hours)**

**Answer: D** By default VACUUM keeps unreferenced files for 7 days so time travel and concurrent readers keep working.

QUESTION 55

**Which command shows a Delta table's location, number of files and size?**

A. DESCRIBE HISTORY

B. SHOW PARTITIONS

**C. ✓ DESCRIBE DETAIL**

D. EXPLAIN

**Answer: C** DESCRIBE DETAIL returns table-level metadata: format, location, numFiles, sizeInBytes and more.

QUESTION 56

**When does a Delta MERGE fail with a 'multiple source rows matched' error?**

A. When WHEN NOT MATCHED is missing

**B. ✓ When more than one source row matches the same target row**

C. When the target table is empty

D. When the source has new columns

**Answer: B** Each target row may be modified by at most one source row. Deduplicate the source first.

QUESTION 57

**Why prefer a SQL UDF ( `CREATE FUNCTION ... RETURN ...` ) to a Python UDF for simple logic?**

A. Python UDFs are always faster

B. SQL UDFs only work on single-node clusters

C. SQL UDFs can't be shared

**D. ✓ SQL UDFs are governed in Unity Catalog, reusable, and can be optimized by the engine without Python serialization**

**Answer: D** SQL UDFs are inlined into the query plan. Python UDFs move every row to a Python process.

## Productionizing Data Pipelines

Medallion architecture, Structured Streaming, Lakeflow Declarative Pipelines, Jobs and Asset Bundles.

**QUESTION 58**

**A streaming job should process all data available right now and then stop, so it can run on a schedule. Which trigger should it use?**

A. processingTime='1 minute'

B. continuous='1 second'

C. No trigger

**D. ✓ availableNow=True**

**Answer: D** availableNow processes everything available (in one or more batches) and then stops. It replaces the deprecated once=True.

**QUESTION 59**

**In a Lakeflow Declarative Pipeline, which expectation drops invalid rows but lets the pipeline continue?**

A. CHECK (id IS NOT NULL)

**B. ✓ EXPECT (id IS NOT NULL) ON VIOLATION DROP ROW**

C. EXPECT (id IS NOT NULL)

D. EXPECT (id IS NOT NULL) ON VIOLATION FAIL UPDATE

**Answer: B** DROP ROW removes failing rows and records them in metrics. Without an ON VIOLATION clause, the rows are kept. FAIL UPDATE stops the update.

**QUESTION 60**

**Which medallion layer typically holds cleaned, deduplicated and conformed data?**

**A. ✓ Silver**

B. Gold

C. Landing

D. Bronze

**Answer: A** Bronze is raw data, silver is cleaned and conformed data, and gold is business-level aggregates.

**QUESTION 61**

**Task 7 of a 10-task job failed because of a bug that is now fixed. How do you finish the run without re-running the tasks that already succeeded?**

A. Delete the failed run

B. Clone the job

C. Start a new run of the whole job

**D. ✓ Use Repair run**

**Answer: D** Repair run re-executes only the failed tasks and the tasks that depend on them.

QUESTION 62

**What is the role of the checkpoint location in Structured Streaming?**

A. It caches the source files

B. It stores the output data

C. It holds the cluster logs

**D. ✓ It stores stream progress and state, which enables fault tolerance and exactly-once processing**

**Answer: D** The checkpoint records offsets and state, so a restarted stream continues where it left off. Each stream needs its own checkpoint.

QUESTION 63

**In a declarative pipeline, which dataset type fits a gold-layer aggregate that should always reflect its source tables?**

A. Temporary view in a notebook

**B. ✓ Materialized view**

C. Streaming table

D. External table

**Answer: B** Materialized views keep their results up to date with the source and are refreshed incrementally when possible. Streaming tables suit append-style incremental ingestion.

QUESTION 64

**What is the recommended way to define jobs and pipelines as code and deploy them to dev and prod workspaces?**

A. Export notebooks as HTML

**B. ✓ Databricks Asset Bundles (databricks.yml plus databricks bundle deploy)**

C. Copy jobs manually in the UI

D. Cluster init scripts

**Answer: B** Asset Bundles describe resources in YAML, use targets for different environments, and are deployed with the Databricks CLI, typically from CI/CD.

QUESTION 65

**A pipeline must keep the full history of customer address changes from a CDC feed. Which option should it use?**

A. INSERT OVERWRITE

**B. ✓ APPLY CHANGES / AUTO CDC ... STORED AS SCD TYPE 2**

C. APPLY CHANGES / AUTO CDC ... STORED AS SCD TYPE 1

D. A materialized view with DISTINCT

**Answer: B** SCD Type 2 keeps a new row for each change, with validity columns. Type 1 overwrites the row in place.

QUESTION 66

**Which pipeline mode reuses the cluster and turns off automatic retries so you can iterate faster?**

A. Production mode

B. Serverless mode

C. Continuous mode

**D. ✓ Development mode**

> **Answer: D** Development mode reuses the cluster and doesn't retry, which helps debugging. Production mode restarts and retries on recoverable failures.

QUESTION 67

**Which Structured Streaming output mode rewrites the entire result table on every trigger?**

**A. ✓ complete**

B. append

C. update

D. overwrite

> **Answer: A** complete mode outputs the full aggregated result each time. append writes only new rows, and update only changed rows.

QUESTION 68

**In a declarative pipeline, what happens to rows that fail `CONSTRAINT c EXPECT (amount > 0)` with no ON VIOLATION clause?**

A. They're dropped

**B. ✓ They're kept, and the violation is recorded in data-quality metrics**

C. They're moved to a quarantine table automatically

D. The update fails

> **Answer: B** With no ON VIOLATION clause the expectation only warns: rows are kept and counted in the metrics.

QUESTION 69

**Where can you see data-quality metrics for a Lakeflow Declarative Pipeline?**

A. Only in the Spark UI

**B. ✓ In the pipeline UI and the pipeline's event log**

C. In DESCRIBE DETAIL

D. They aren't recorded

> **Answer: B** Expectation results are recorded in the event log and shown in the pipeline UI for each dataset.

QUESTION 70

**How can one task in a Lakeflow Job pass a small value (for example a row count) to a downstream task?**

**A. ✓ Use task values (dbutils.jobs.taskValues.set / get)**

B. Store it in the cluster's Spark config

C. Email it

D. Write it to a global variable

**Answer: A** Task values pass small pieces of data between tasks in the same job run.

QUESTION 71

**Which trigger starts a job when new files land in a storage location?**

A. Cron schedule

B. Continuous trigger

C. Manual trigger

**D. ✓ File arrival trigger**

**Answer: D** File arrival triggers watch a Unity Catalog location and start the job when new files appear.

QUESTION 72

**A Delta table used as a streaming source receives UPDATE and DELETE operations. What happens by default?**

A. Nothing changes

**B. ✓ The stream fails, because streaming sources must be append-only (unless you set options such as skipChangeCommits)**

C. The table is converted to append-only

D. The stream processes them as new rows

**Answer: B** Streaming from Delta assumes appends. Use skipChangeCommits or the change data feed to handle updates.

QUESTION 73

**Which medallion layer holds business-level aggregates used by dashboards?**

A. Bronze

**B. ✓ Gold**

C. Landing

D. Silver

**Answer: B** Gold tables are modelled for consumption: aggregates, star schemas and features.

QUESTION 74

**How does a triggered Lakeflow Declarative Pipeline differ from a continuous one?**

A. A continuous pipeline only runs once a day

B. A triggered pipeline never stops

**C. ✓ A triggered pipeline processes all available data and then stops**

D. There's no difference

**Answer: C** Triggered mode processes the backlog and shuts down. Continuous mode keeps running for low latency.

QUESTION 75

**Which command deploys a Databricks Asset Bundle to its prod target?**

**A. ✓ databricks bundle deploy -t prod**

B. databricks workspace import prod

C. git push prod

D. databricks jobs create prod

> **Answer: A** bundle deploy creates or updates the bundle's resources in the target workspace. Then use bundle run to start a job.

## Data Governance & Quality

Unity Catalog, privileges, managed vs external tables, lineage and Delta Sharing.

QUESTION 76

**What is the correct three-level namespace in Unity Catalog?**

**A. ✓ catalog.schema.table**

B. workspace.schema.table

C. metastore.catalog.table

D. database.schema.table

> **Answer: A** Objects are addressed as catalog.schema.object. The metastore is the container above catalogs.

QUESTION 77

**You run DROP TABLE on an external table. What happens to the underlying data files?**

A. They are moved to a trash folder

**B. ✓ They remain in storage. Only the metadata is removed.**

C. They are deleted immediately

D. They are converted to a managed table

> **Answer: B** For external tables, Unity Catalog doesn't manage the files, so DROP TABLE removes only the table definition.

QUESTION 78

**Group `analysts` has SELECT on main.sales.orders but gets a permission error when querying it. What is most likely missing?**

A. MODIFY on the table

**B. ✓ USE CATALOG on main and USE SCHEMA on main.sales**

C. CREATE TABLE on the schema

D. Ownership of the table

> **Answer: B** Reading a table needs USE CATALOG and USE SCHEMA on its parent objects in addition to SELECT on the table.

QUESTION 79

**Which statement grants read access on every current and future table in schema main.sales?**

**A. ✓ GRANT SELECT ON SCHEMA main.sales TO** `analysts`

B. GRANT READ ON main.sales TO `analysts`

C. ALTER SCHEMA main.sales OWNER TO `analysts`

D. GRANT SELECT ON ALL TABLES main.sales TO `analysts`

**Answer: A** Unity Catalog privileges inherit downwards, so SELECT on a schema applies to all tables in it, including ones created later.

QUESTION 80

**A company wants to share live Delta tables with a partner who uses pandas outside Databricks. What should it use?**

A. Export CSVs nightly to email

B. Lakehouse Federation

C. Databricks-to-Databricks Delta Sharing

**D. ✓ Open Delta Sharing with a recipient credential**

**Answer: D** Open sharing lets recipients on any platform read shared data using the open Delta Sharing protocol.

QUESTION 81

**Where can you see which upstream tables and columns a gold table was derived from?**

A. The Spark UI

**B. ✓ Unity Catalog lineage in Catalog Explorer**

C. The cluster driver logs

D. DESCRIBE DETAIL

**Answer: B** Unity Catalog automatically captures table and column lineage, which you can view in Catalog Explorer and query through system tables.

QUESTION 82

**What is required before you can create an external table at a cloud storage path in Unity Catalog?**

**A. ✓ An external location (a storage credential plus a path) you have privileges on**

B. A DBFS mount

C. A cluster pool

D. A Delta Share

**Answer: A** External locations pair a storage credential with a path and govern access to that path.

QUESTION 83

**Which Unity Catalog feature shows masked email addresses to users who aren't in the `pii_readers` group?**

A. Row filter

**B. ✓ Column mask**

C. Delta Sharing

D. Table ACL on the hive_metastore

> **Answer: B** A column mask is a UC function attached to a column that returns either the real value or a masked one, for example based on is_account_group_member().

QUESTION 84

**Which UC object governs non-tabular files, such as images or raw CSVs, at paths like /Volumes/main/raw/landing?**

**A. ✓ Volume**

B. External table

C. View

D. Share

> **Answer: A** Volumes are Unity Catalog objects for files. Access is controlled with READ VOLUME and WRITE VOLUME.

QUESTION 85

**Besides USE CATALOG and USE SCHEMA, which privilege does a user need to create tables in schema main.sales?**

A. MODIFY on the catalog

**B. ✓ CREATE TABLE on the schema**

C. SELECT on the schema

D. ALL PRIVILEGES on the metastore

> **Answer: B** CREATE TABLE on the schema (with USE CATALOG and USE SCHEMA) allows creating tables in it.

QUESTION 86

**Who can grant privileges on a Unity Catalog table?**

**A. ✓ The table's owner, users with MANAGE on it, and admins of its parent objects**

B. Anyone with SELECT

C. Only the workspace admin

D. Only Databricks support

> **Answer: A** Owners (and principals with MANAGE, catalog owners and metastore admins) can grant access.

QUESTION 87

**How is a row filter implemented in Unity Catalog?**

A. As a view that every user must query

B. As a cluster setting

C. As a Delta table property

**D. ✓ As a SQL UDF attached with ALTER TABLE ... SET ROW FILTER**

**Answer: D** The filter function returns a boolean per row and is applied automatically to every query on the table.

QUESTION 88

**What can a Delta Sharing recipient do with shared tables?**

A. Delete them

B. Read and write

C. Change their schema

**D. ✓ Read them only**

**Answer: D** Shares are read-only for recipients. The provider stays in control and can revoke access at any time.

QUESTION 89

**Which system table holds audit events for workspaces and the account?**

A. information_schema.audit

**B. ✓ system.access.audit**

C. system.billing.usage

D. system.compute.clusters

**Answer: B** system.access.audit records actions such as logins, permission changes and data access.

QUESTION 90

**Where are a managed table's data files stored?**

**A. ✓ In the managed storage location of its schema, catalog or metastore, managed by Unity Catalog**

B. In the control plane

C. Wherever the user chooses with LOCATION

D. On the driver's disk

**Answer: A** Unity Catalog manages the location and lifecycle of managed tables' files.

QUESTION 91

**In a Unity Catalog-enabled workspace, how does the legacy Hive metastore appear?**

A. As a schema inside main

**B. ✓ As a catalog named hive_metastore**

C. It disappears

D. As a Delta Share

**Answer: B** Legacy tables stay reachable through the hive_metastore catalog until they're upgraded.

QUESTION 92

**Which function is used in row filters, column masks and dynamic views to check a user's group?**

A. has_role('group')

B. user_in('group')

**C. ✓ is_account_group_member('group')**

D. current_group()

> **Answer: C** is_account_group_member checks account-level group membership of the current user.

QUESTION 93

**Which statement removes a group's read access to a table?**

**A. ✓ REVOKE SELECT ON TABLE t FROM** `analysts`

B. DELETE GRANT SELECT ON t FROM `analysts`

C. DROP PRIVILEGE SELECT ON t

D. DENY SELECT ON t TO `analysts`

> **Answer: A** REVOKE undoes a GRANT. Unity Catalog has no DENY.

# PySpark track

## DataFrame Fundamentals

SparkSession, lazy evaluation, columns and conditions, null handling, aggregations and joins.

**QUESTION 94**

**Which of these PySpark calls is an action that triggers execution?**

A. df.withColumn('x', F.lit(1))

B. df.select('a')

C. df.filter(F.col('a') > 1)

**D. ✓ df.count()**

> **Answer: D** count() returns a number to the driver, so Spark has to run the plan. The others are lazy transformations that return a new DataFrame.

**QUESTION 95**

**`df.filter(F.col('a') > 1 and F.col('b') < 5)` raises an error. What's the fix?**

A. Use `df.where` instead of `df.filter`

**B. ✓ Use `(F.col('a') > 1) & (F.col('b') < 5)`**

C. Use `F.col('a') > 1 & F.col('b') < 5` without parentheses

D. Cast both columns to boolean

> **Answer: B** Python's `and` tries to turn a Column into a bool. Use `&`, `|` and `~`, with each condition in parentheses because `&` binds more tightly than comparison operators.

**QUESTION 96**

**What does `F.when(F.col('price') > 100, 'high')` return for rows where price is 50, if there's no `.otherwise()`?**

A. 'low'

**B. ✓ NULL**

C. The original price

D. An error

> **Answer: B** Without otherwise(), rows that match no condition get NULL.

**QUESTION 97**

**How do `a.join(b, 'id')` and `a.join(b, a.id == b.id)` differ?**

A. The string form does a left join

B. The expression form only works with broadcast joins

**C. ✓ The string form keeps a single `id` column. The expression form keeps both, so a bare `id` becomes ambiguous.**

D. They are identical

> **Answer: C** Joining on column names (USING semantics) merges the key into one column. Joining on an expression keeps a.id and b.id.

QUESTION 98

**Which join returns customers who have never placed an order, with only the customers' columns?**

A. customers.join(orders, 'customer_id', 'outer')

**B. ✓ customers.join(orders, 'customer_id', 'left_anti')**

C. customers.join(orders, 'customer_id', 'left')

D. customers.join(orders, 'customer_id', 'left_semi')

> **Answer: B** left_anti keeps left rows without a match. left_semi keeps left rows that do have a match.

QUESTION 99

**`df.fillna(0)` is run on a DataFrame with an integer column `qty` and a string column `country`, both with NULLs. What happens?**

A. It fails because of the string column

**B. ✓ Only `qty` NULLs become 0. `country` is unchanged.**

C. `country` NULLs become '0'

D. Both columns get 0

> **Answer: B** A scalar fill value only applies to columns of a compatible type. Use a dict, as in fillna({'country': 'UNKNOWN'}), for per-column values.

QUESTION 100

**What is the name of the column created by `df.groupBy('status').count()`?**

A. status_count

B. count(1)

**C. ✓ count**

D. n

> **Answer: C** GroupedData.count() produces a column named `count`. F.count('*') inside agg() would be named `count(1)` unless you alias it.

QUESTION 101

**What does `df.withColumn('price', F.col('price') * 2)` do when `price` already exists?**

A. Adds a second column named price

B. Raises an error

C. Creates price_2

**D. ✓ Replaces the existing price column**

> **Answer: D** withColumn with an existing name replaces that column in place.

QUESTION 102

**What is `F.lit()` for?**

A. Lowercasing text

B. Limiting rows

**C. ✓ Turning a Python value into a Column, for example F.lit('UNKNOWN')**

D. Reading a literal file

> **Answer: C** Functions treat strings as column names, so wrap constants in F.lit() when you need a literal.

QUESTION 103

**`df.dropna(how='all', subset=['email', 'phone'])` drops which rows?**

**A. ✓ Rows where both email and phone are NULL**

B. Rows where email or phone is NULL

C. No rows

D. Rows where any column is NULL

**Answer: A** how='all' drops a row only when every listed column is NULL. how='any' (the default) drops it if any of them is.

QUESTION 104

**How does `df1.union(df2)` match columns?**

A. By data type

B. By name

C. Randomly

**D. ✓ By position**

**Answer: D** union is positional. Use unionByName to match by column name.

QUESTION 105

**Why use `orders.alias('o')` before a join?**

A. To cache the DataFrame

**B. ✓ So columns can be referenced as F.col('o.customer_id'), for example in self-joins**

C. To change the join type

D. To rename the table permanently

**Answer: B** DataFrame aliases qualify column references, which removes ambiguity.

QUESTION 106

**With ANSI mode off, what does `F.col('x').cast('int')` return for the string 'abc'?**

A. An error

B. 0

**C. ✓ NULL**

D. 'abc'

**Answer: C** Without ANSI mode, invalid casts return NULL. With ANSI mode on (the default in newer runtimes), they raise an error. try_cast always returns NULL.

QUESTION 107

**Which code counts the distinct statuses in `orders`?**

**A. ✓ orders.select('status').distinct().count()**

B. orders.distinct('status')

C. orders.count('status')

D. orders.status.count()

**Answer: A** Select the column, take distinct values and count them. F.countDistinct('status') inside agg() works too.

## Window & Advanced Functions

Ranking, running totals and lag/lead, arrays and explode, structs, higher-order functions, dates.

QUESTION 108

**Three products in one category cost 100, 100 and 90. What does `F.dense_rank()` ordered by price descending return?**

A. 1, 2, 3

B. 2, 2, 1

C. 1, 1, 3

**D. ✓ 1, 1, 2**

> **Answer: D** dense_rank gives ties the same rank with no gaps. rank would give 1, 1, 3 and row_number 1, 2, 3.

QUESTION 109

**What does `F.sum('amount').over(Window.partitionBy('customer_id').orderBy('order_date'))` compute?**

A. An error, because a frame is required

B. The amount of the previous order

C. The customer's total over all orders, on every row

**D. ✓ A running total up to and including the current order date**

> **Answer: D** With an orderBy and no explicit frame, the frame runs from the start of the partition to the current row (including peers with the same date). Without orderBy it would be the whole partition.

QUESTION 110

**A row's `items` array is empty. What happens to that row with `F.explode('items')`?**

**A. ✓ It's removed from the result**

B. It becomes one row with an empty string

C. It stays, with a NULL item

D. It raises an error

> **Answer: A** explode drops rows whose array is empty or NULL. explode_outer keeps them with a NULL element.

QUESTION 111

**What does `F.lag('order_date', 1)` return for the first row in each window partition?**

A. The last row of the previous partition

**B. ✓ NULL (unless a default is given)**

C. The same order_date

D. 0

> **Answer: B** There's no previous row, so lag returns NULL, or the third argument if you pass a default.

QUESTION 112

**How do `collect_list` and `collect_set` differ?**

A. collect_set keeps duplicates

B. collect_list removes duplicates

C. There's no difference

**D. ✓ collect_set removes duplicates and doesn't guarantee order**

> **Answer: D** collect_set returns distinct values with no guaranteed order. Wrap it in array_sort for a stable result. collect_list keeps every value.

QUESTION 113

**Which date pattern formats 2024-03-07 as `2024-03`?**

A. '%Y-%m'

B. 'YYYY-MM'

C. 'yyyy-mm'

**D. ✓ 'yyyy-MM'**

> **Answer: D** Spark uses Java patterns: MM is the month and mm is minutes. 'YYYY' is the week-based year, and '%Y' is Python/C strftime syntax.

QUESTION 114

**What does `F.sum('amount').over(Window.partitionBy('customer_id'))` (with no orderBy) compute?**

A. The previous amount

B. A running total

**C. ✓ The customer's total amount, on every row of that customer**

D. The overall total for the DataFrame

> **Answer: C** Without orderBy the frame is the whole partition, so every row gets the partition total.

QUESTION 115

**Which rows does `rowsBetween(-2, 0)` include?**

A. The whole partition

B. Only the two preceding rows

C. The current row and the two following rows

**D. ✓ The current row and the two preceding rows**

> **Answer: D** Negative offsets are preceding rows and 0 is the current row, so this is a 3-row moving window.

QUESTION 116

**What does `F.transform('prices', lambda p: p * 1.2)` return?**

A. The sum of the prices

B. One row per price

**C. ✓ A new array with every price multiplied by 1.2**

D. A filtered array

> **Answer: C** transform maps a lambda over each array element and keeps the array shape.

QUESTION 117

**What is the type of `F.split(F.col('tags'), ',')`?**

**A. ✓ array<string>**

B. map<string,string>

C. struct

D. string

> **Answer: A** split returns an array of strings. Use getItem(i) or explode to work with its elements.

QUESTION 118

**Which call returns the number of days from `start_date` to `end_date`?**

A. F.months_between('start_date', 'end_date')

B. F.date_sub('end_date', 'start_date')

**C. ✓ F.datediff('end_date', 'start_date')**

D. F.datediff('start_date', 'end_date')

**Answer: C** datediff(end, start): the end date comes first.

QUESTION 119

**How do you select the `city` field of a struct column `address`?**

A. F.split('address', 'city')

**B. ✓ F.col('address')['city'] or F.col('address.city')**

C. F.explode('address')

D. address->city

**Answer: B** Struct fields are accessed with dot notation, getField('city') or ['city'].

## Reading, Writing & Delta

DataFrameReader and schemas, save modes and partitioning, the DeltaTable API and time travel.

QUESTION 120

**What is the default save mode of `df.write.saveAsTable('t')` when table t already exists?**

A. overwrite

B. ignore

**C. ✓ errorifexists (the write fails)**

D. append

**Answer: C** The default mode is errorifexists. Set mode('append') or mode('overwrite') explicitly.

QUESTION 121

**Why provide an explicit schema when reading large CSV or JSON files?**

A. It enables time travel

B. It compresses the data

**C. ✓ It avoids an extra inference pass over the data and prevents wrongly inferred types**

D. It's required for CSV

**Answer: C** Schema inference reads data just to guess types, and it can guess wrong (for example, dropping leading zeros from IDs). An explicit schema is faster and predictable.

QUESTION 122

**You build a DeltaTable merge with `.merge(...).whenMatchedUpdateAll().whenNotMatchedInsertAll()` but the table doesn't change. What's missing?**

A. spark.sql('COMMIT')

B. .commit()

**C. ✓ .execute()**

D. .save()

> **Answer: C** The merge builder does nothing until you call .execute().

QUESTION 123

**Which is a good use of `.partitionBy(...)` when writing a large Delta table?**

A. The primary key, to make updates faster

**B. ✓ A low-cardinality column that most queries filter on, such as country or event_date**

C. Every column, for maximum speed

D. customer_id, with millions of values

> **Answer: B** Partitioning creates a directory per value, so high cardinality creates huge numbers of small files. For new tables, liquid clustering is often the better choice.

QUESTION 124

**How do you read version 3 of a Delta table from Python?**

A. DeltaTable.forName(spark, 't').version(3)

B. spark.read.version(3).table('t')

**C. ✓ spark.read.option('versionAsOf', 3).table('t')**

D. spark.table('t', version=3)

> **Answer: C** Use the versionAsOf (or timestampAsOf) reader option. In SQL: SELECT * FROM t VERSION AS OF 3.

QUESTION 125

**An append adds a new column that the target Delta table doesn't have. What lets the write succeed?**

**A. ✓ .option('mergeSchema', 'true')**

B. .mode('ignore')

C. Nothing: Delta always accepts new columns

D. .option('inferSchema', 'true')

> **Answer: A** Delta enforces the schema by default. mergeSchema allows schema evolution (adding columns) on write.

QUESTION 126

**You read a CSV file with a header row but forget `.option('header', 'true')`. What happens?**

A. The first column is dropped

B. The read fails

C. Spark detects the header automatically

**D. ✓ The header line is read as a data row and columns are named _c0, _c1, ...**

> **Answer: D** header defaults to false for CSV.

QUESTION 127

**Without a schema and without `inferSchema`, what type do CSV columns get?**

A. binary

B. NULL

C. They're inferred automatically

**D. ✓ string**

> **Answer: D** inferSchema is off by default for CSV, so every column is a string.

QUESTION 128

**Which reader mode makes the read fail as soon as a malformed record is found?**

A. STRICT

B. PERMISSIVE

C. DROPMALFORMED

**D. ✓ FAILFAST**

> **Answer: D** FAILFAST stops at the first bad record. PERMISSIVE (the default) sets bad fields to NULL, and DROPMALFORMED skips bad rows.

QUESTION 129

**What does `.option('overwriteSchema', 'true')` allow on a Delta write with mode('overwrite')?**

**A. ✓ Replacing the table's schema (for example changing column types) along with the data**

B. Writing to a different table

C. Appending rows

D. Skipping schema checks on reads

> **Answer: A** A normal overwrite must keep the schema. overwriteSchema lets you change it.

QUESTION 130

**What does `DeltaTable.forName(spark, 't').history()` return?**

A. A list of file paths

B. The current data

C. A string

**D. ✓ A DataFrame with a row per table version: version, timestamp, operation, user and metrics**

> **Answer: D** history() is the Python equivalent of DESCRIBE HISTORY and returns a DataFrame.

QUESTION 131

**Which reader option reads row-level changes from a Delta table that has change data feed enabled?**

**A. ✓ readChangeFeed**

B. versionAsOf

C. ignoreChanges

D. mergeSchema

> **Answer: A** spark.read.option('readChangeFeed', 'true').option('startingVersion', n).table(...) returns the changes with _change_type metadata.

## Performance & Internals

How Spark plans and runs a query: shuffles, partitions, AQE, broadcast joins, caching and UDF cost.

QUESTION 132

**Which transformation causes a shuffle?**

A. withColumn

B. select

**C. ✓ groupBy(...).agg(...)**

D. filter

**Answer: C** Aggregating by key needs all rows with the same key in the same partition, which means a wide transformation and a shuffle. filter, withColumn and select are narrow.

QUESTION 133

**When is a broadcast join a good choice?**

**A. ✓ When one side is small enough to copy to every executor**

B. When both tables are huge

C. Only for full outer joins

D. When the join key has many NULLs

**Answer: A** Broadcasting the small side avoids shuffling the large side. Broadcasting a large table can cause out-of-memory errors.

QUESTION 134

**You call `df.cache()`. When is the data actually cached?**

**A. ✓ When the first action computes df**

B. Immediately

C. At the end of the notebook

D. When you call df.persist()

**Answer: A** cache() is lazy. The data is materialized the first time an action computes it.

QUESTION 135

**Why is a row-at-a-time Python UDF usually slower than built-in functions?**

A. UDFs can't run in parallel

B. Python is always single-threaded on Spark

C. UDFs disable Photon for the whole cluster

**D. ✓ Each row is serialized between the JVM and a Python worker, and the UDF is opaque to the optimizer**

**Answer: D** Row-by-row serialization is expensive, and Catalyst can't optimize inside a UDF. Built-ins are best. pandas_udf (Arrow batches) is the faster option when you need custom Python.

QUESTION 136

**Which of these does Adaptive Query Execution (AQE) do at runtime?**

**A. ✓ Coalesces small shuffle partitions, switches join strategies and splits skewed partitions**

B. Automatically adds indexes

C. Caches every DataFrame

D. Rewrites Python code into SQL

**Answer: A** AQE re-optimizes using statistics from completed stages: partition coalescing, dynamic join strategy changes and skew-join handling.

QUESTION 137

**What is the default value of `spark.sql.shuffle.partitions`?**

A. The number of cores

**B. ✓ 200**

C. 64

D. 8

**Answer: B** It's 200. AQE usually coalesces the resulting partitions to a sensible number.

QUESTION 138

**In the Spark UI, most tasks in a join stage finish in 10 seconds, but three take 15 minutes. What is the likely cause?**

A. Too many executors

B. The driver is out of memory

**C. ✓ Data skew: a few keys have far more rows than others**

D. The broadcast threshold is too low

**Answer: C** A few very slow tasks in an otherwise fast stage is the classic sign of skew. AQE skew handling, salting or broadcasting the small side help.

QUESTION 139

**How do `df.coalesce(1)` and `df.repartition(1)` differ?**

**A. ✓ coalesce merges partitions without a full shuffle; repartition does a full shuffle**

B. They're identical

C. repartition never shuffles

D. coalesce increases partitions

**Answer: A** coalesce avoids a shuffle but can reduce upstream parallelism. repartition shuffles to get evenly sized partitions.

QUESTION 140

**What is the default value of `spark.sql.autoBroadcastJoinThreshold`?**

A. 1 GB

**B. ✓ 10 MB**

C. No limit

D. 1 MB

**Answer: B** Tables smaller than about 10 MB are broadcast automatically. AQE can also switch to a broadcast join at runtime.

QUESTION 141

**What does predicate pushdown do?**

A. Moves data between regions

B. Pushes results to the driver

**C. ✓ Applies filters at the data source, so less data is read**

D. Pushes code to the executors

> **Answer: C** Catalyst pushes filters into the scan. With Delta statistics, files that can't match are skipped entirely.

QUESTION 142

**How many tasks process one partition in a stage?**

**A. ✓ One task per partition**

B. One task per row

C. One task per column

D. One task per executor

> **Answer: A** Spark schedules a task for each partition of a stage, so the partition count sets the stage's parallelism.

QUESTION 143

**Why is a `pandas_udf` usually faster than a row-at-a-time `udf`?**

A. It runs on the driver

B. It uses SQL internally

**C. ✓ It processes data in vectorized Arrow batches instead of one row at a time**

D. It skips data

> **Answer: C** Arrow batches make serialization much cheaper, and pandas operations are vectorized.

QUESTION 144

**What does `df.explain()` show?**

**A. ✓ The physical plan Spark will run**

B. The Spark UI

C. The data

D. The table history

> **Answer: A** explain() prints the plan. explain('formatted') or explain(True) also shows the logical plans.

QUESTION 145

**What does 'spill' in the Spark UI mean?**

A. A broadcast happened

B. The cluster autoscaled

C. Data was lost

**D. ✓ Data didn't fit in execution memory during a shuffle or sort and was written to disk**

> **Answer: D** Spill slows stages down. More or larger partitions, more memory per core, or avoiding skew all reduce it.

# Solutions Architect track (Azure)

## Azure Databricks Platform Architecture

Control and compute planes, workspace deployment, VNet injection, secure cluster connectivity, Private Link and serverless networking.

QUESTION 146

**What does VNet injection require for each Azure Databricks workspace?**

**A. ✓ Two dedicated subnets (host and container), delegated to Microsoft.Databricks/workspaces, each with an NSG**

B. An ExpressRoute circuit

C. One subnet shared with other Azure services

D. A public IP for every cluster node

> **Answer: A** Each workspace needs its own host and container subnets, delegated to Databricks and associated with network security groups. Every cluster node uses one IP in each.

QUESTION 147

**With secure cluster connectivity (No Public IP), how does the control plane manage cluster VMs?**

A. Through a public load balancer in the customer VNet

**B. ✓ Each cluster opens an outbound connection to the SCC relay, and no inbound ports are open**

C. Through Azure Bastion

D. Through inbound SSH from the control plane

> **Answer: B** Cluster nodes have no public IPs and accept no inbound traffic. They initiate an outbound TLS tunnel to the relay.

QUESTION 148

**A bank wants users to reach the workspace UI and REST API only from its corporate network, never over the public internet. What does that need?**

A. Larger subnets

B. A NAT gateway

C. Back-end Private Link only

**D. ✓ Front-end Private Link (plus a browser-authentication endpoint), with public network access disabled**

> **Answer: D** Front-end (user-to-workspace) Private Link covers user and API traffic. Web SSO also needs the browser-authentication private endpoint for the region.

QUESTION 149

**The storage account firewall blocks serverless SQL warehouses from reading ADLS. What's the correct fix?**

A. Enable public access on the storage account

B. Add the workspace VNet's subnets to the storage firewall

C. Switch the warehouse to classic

**D. ✓ Attach a network connectivity configuration (NCC) and use its private endpoints or stable subnets for the storage account**

> **Answer: D** Serverless compute doesn't run in your VNet. An NCC gives it private endpoints, or subnet IDs you can allow, for your storage.

QUESTION 150

**Which component runs in the Databricks-managed control plane rather than your Azure subscription?**

A. The NAT gateway

B. Classic cluster VMs

C. The workspace's ADLS data

**D. ✓ The web application, job scheduler and cluster manager**

> **Answer: D** The control plane hosts the web app, APIs, scheduler and cluster manager. Classic VMs and your data live in your subscription.

QUESTION 151

**Why is getting subnet size right at workspace creation important?**

A. Small subnets disable Unity Catalog

**B. ✓ Subnet size limits how many cluster nodes the workspace can run at once, and is hard to change later**

C. Subnet size sets the Spark version

D. Bigger subnets reduce DBU prices

> **Answer: B** Every node takes one IP in each subnet, so the smaller subnet caps total concurrent nodes across all clusters in the workspace.

QUESTION 152

**With secure cluster connectivity, which inbound NSG rules do the workspace subnets need from the internet?**

**A. ✓ None**

B. SSH (22) from the control plane

C. All ports from Databricks IPs

D. HTTPS (443) from anywhere

> **Answer: A** With SCC, clusters only make outbound connections, so no inbound rules from the internet are needed.

QUESTION 153

**Can two Azure Databricks workspaces share the same host and container subnets?**

A. Only in the same resource group

**B. ✓ No, each workspace needs its own dedicated pair of subnets**

C. Yes, always

D. Only on serverless

**Answer: B** Subnets are delegated to one workspace. Plan separate subnet pairs (they can be in the same VNet).

QUESTION 154

**Clusters in a VNet-injected workspace with SCC have no public IPs. What gives them outbound connectivity?**

A. Their public IPs

B. No outbound connectivity is needed

**C. ✓ A NAT gateway, or a route (UDR) to Azure Firewall or an NVA**

D. The control plane proxies all traffic

**Answer: C** Something has to provide egress, typically a NAT gateway or a firewall route. Back-end Private Link keeps control-plane traffic private.

QUESTION 155

**Which pricing tier is required for Private Link and Unity Catalog on Azure Databricks?**

**A. ✓ Premium**

B. Standard

C. Any tier

D. Trial

**Answer: A** Enterprise security and governance features need the Premium tier.

QUESTION 156

**For data-residency reviews, where does Azure Databricks store your Unity Catalog table data?**

A. In Databricks' control plane

**B. ✓ In your ADLS Gen2 storage accounts, in the regions you choose**

C. In a global Databricks bucket

D. On cluster local disks

**Answer: B** Table data stays in your storage. The control plane holds metadata and workspace assets.

QUESTION 157

**Where does serverless compute run?**

A. In your subscription's VNet

B. In another cloud

**C. ✓ In a Databricks-managed serverless compute plane in the same region as the workspace**

D. On-premises

**Answer: C** Serverless runs in Databricks' account, in the workspace's region, with network isolation between customers.

## Identity, Security & Governance Design

Entra ID and account-level identity, service principals, secrets, encryption keys and Unity Catalog topology.

**QUESTION 158**

**What is the recommended way to manage users and groups for Azure Databricks?**

A. Create users manually in each workspace

B. Use workspace-local groups for Unity Catalog grants

**C. ✓ Sync from Entra ID to the Databricks account (automatic identity management or SCIM) and assign account groups to workspaces**

D. Share one admin login

> **Answer: C** Account-level identities from Entra ID can be used everywhere, including in Unity Catalog grants. Workspace-local groups can't.

**QUESTION 159**

**How should Unity Catalog access ADLS Gen2 on Azure?**

A. A SAS token in each notebook

**B. ✓ An Access Connector for Azure Databricks (managed identity) registered as a storage credential, with external locations**

C. Mount points with a personal service principal secret

D. Storage account keys in Spark config

> **Answer: B** A managed identity has no secret to leak or rotate. UC governs the paths through external locations, and users never get storage permissions.

**QUESTION 160**

**Which identity should run production jobs and CI/CD deployments?**

A. The metastore admin

B. The lead engineer's account

C. A shared user account with a PAT

**D. ✓ A service principal using OAuth or Entra ID tokens**

> **Answer: D** Service principals decouple production from people. They use short-lived OAuth or Entra tokens instead of long-lived PATs.

**QUESTION 161**

**Dev and prod workspaces share one metastore. How do you make sure production catalogs can't be accessed from the dev workspace, even by users who have grants?**

**A. ✓ Workspace-catalog binding: bind prod catalogs only to prod workspaces**

B. Use a different storage account

C. Create a second metastore in the same region

D. Rename the catalogs

> **Answer: A** Bindings restrict which workspaces can use a catalog. Grants decide who can use it. You can only have one metastore per region.

QUESTION 162

**How many Unity Catalog metastores should an account have in one Azure region?**

A. One per workspace

**B. ✓ One per region, shared by the workspaces there**

C. One per business unit

D. As many as you like, for isolation

**Answer: B** A metastore is regional and shared by every workspace in that region. Isolate with catalogs, bindings and grants.

QUESTION 163

**Which customer-managed key option protects notebook source and secrets stored in the control plane?**

**A. ✓ Managed services**

B. ADLS encryption

C. Managed disks

D. Workspace storage

**Answer: A** The managed services key covers control-plane data at rest. Managed disks covers cluster VM disks, and workspace storage covers the DBFS root account.

QUESTION 164

**A notebook prints `dbutils.secrets.get('kv', 'db-password')`. What appears in the output?**

A. An error

B. The plain password

C. The Key Vault URL

**D. ✓ [REDACTED]**

**Answer: D** Databricks redacts secret values in notebook output. Access to the secret scope still has to be controlled.

QUESTION 165

**Which kind of group can be used in Unity Catalog grants?**

A. Azure resource groups

**B. ✓ Account-level groups (for example, synced from Entra ID)**

C. Workspace-local groups

D. Spark groups

**Answer: B** Unity Catalog grants use account-level identities. Workspace-local groups are a legacy construct.

QUESTION 166

**What is the difference between a storage credential and an external location?**

A. An external location is a table

B. They are the same

**C. ✓ A storage credential is the identity (for example an Access Connector); an external location pairs a storage path with that credential**

D. A storage credential is a SAS token per user

> **Answer: C** You grant access on external locations (READ FILES, CREATE EXTERNAL TABLE ...). The credential stays hidden from users.

QUESTION 167

**How do you give a group read access to all current and future tables in catalog sales_prod?**

A. Make the group the owner of every table

**B. ✓ GRANT USE CATALOG, USE SCHEMA, SELECT ON CATALOG sales_prod TO the group**

C. Bind the catalog to the group

D. Grant SELECT on each table every night

> **Answer: B** Privileges granted on a catalog are inherited by every schema and table in it, including ones created later.

QUESTION 168

**Auditors ask which downstream tables use a salary column. Where do you look?**

A. The Spark UI

**B. ✓ Unity Catalog column-level lineage (Catalog Explorer or the lineage system tables)**

C. Cluster event logs

D. Notebook revision history

> **Answer: B** Unity Catalog captures column-level lineage automatically for queries on UC-enabled compute.

QUESTION 169

**Who should own production catalogs and schemas?**

A. The engineer who created them

**B. ✓ A group or service principal, so ownership survives people leaving**

C. Nobody

D. Every user

> **Answer: B** Group ownership avoids orphaned objects and makes administration shared and auditable.

## Lakehouse Reference Architectures

Batch and streaming ingestion on Azure, medallion pipelines, BI serving with Power BI, and ML/GenAI architectures.

QUESTION 170

**Millions of files a day land in ADLS. Which Auto Loader setup scales best?**

**A. ✓ File notification mode (Event Grid plus a storage queue) or managed file events**

B. A Python script with dbutils.fs.ls

C. Directory listing every minute

D. COPY INTO in a loop

**Answer: A** Notification-based discovery avoids listing huge directories. It's the scalable mode for high file volumes.

QUESTION 171

**How does Spark Structured Streaming typically read from Azure Event Hubs?**

A. Through a JDBC driver

**B. ✓ Through the Event Hubs Kafka-compatible endpoint with the Kafka source**

C. With COPY INTO

D. By polling Event Hubs with REST calls in a UDF

**Answer: B** Event Hubs exposes a Kafka endpoint, so the built-in Kafka source works with SASL configuration.

QUESTION 172

**A Power BI report on a 2-billion-row fact table needs data that's 5 minutes fresh. What's the best pattern?**

A. Query bronze directly

**B. ✓ DirectQuery (or composite with aggregations) against a serverless SQL warehouse, over a gold table**

C. Import mode refreshed every 5 minutes

D. Export CSV files to SharePoint

**Answer: B** DirectQuery keeps data fresh without huge imports. Size the warehouse for report concurrency and model gold for BI.

QUESTION 173

**Which components form a governed RAG architecture on Databricks?**

A. A pandas DataFrame and a Flask app on a VM

B. Delta Sharing and Power BI

C. A SQL warehouse only

**D. ✓ Chunked documents in Delta, a Vector Search index synced from them, and an agent served through Model Serving, all in Unity Catalog**

**Answer: D** Vector Search syncs from Delta tables, and Model Serving hosts the agent. Unity Catalog governs and tracks lineage for both.

QUESTION 174

**When is Lakehouse Federation the right tool?**

A. To replicate data across regions

B. For high-volume nightly ETL from SQL Server

C. To serve real-time ML features

**D. ✓ To query an operational database in place for exploration or migration validation**

> **Answer: D** Federation pushes queries down to the source without copying data. Recurring heavy workloads should ingest into Delta instead.

QUESTION 175

**The business wants dashboards updated every 30 minutes. What is the most cost-effective streaming design?**

A. A manual refresh button

B. A continuous pipeline running 24×7

**C. ✓ A triggered pipeline or availableNow job scheduled every 30 minutes**

D. An all-purpose cluster that never terminates

> **Answer: C** Triggered runs process only new data and then stop, which meets a 30-minute SLA without paying for always-on compute.

QUESTION 176

**What does an Event Hub's partition count limit for a Spark streaming consumer?**

A. The retention period

B. The number of producers

**C. ✓ The maximum read parallelism (concurrent consumers)**

D. The message size

> **Answer: C** Each partition is read by one consumer task at a time. Choose enough partitions for the target throughput.

QUESTION 177

**A pipeline must keep the full history of customer attribute changes from a CDC feed. Which pattern fits?**

A. A materialized view with DISTINCT

B. INSERT OVERWRITE every night

**C. ✓ AUTO CDC (APPLY CHANGES) ... STORED AS SCD TYPE 2 in a declarative pipeline**

D. Delta Sharing

> **Answer: C** SCD Type 2 keeps a version row per change with validity columns, built from CDC events.

QUESTION 178

**How should gold data for Power BI typically be modelled?**

A. As raw JSON

**B. ✓ As star schemas (facts and dimensions) with clear keys**

C. As bronze tables

D. As one very wide CSV

> **Answer: B** Star schemas suit BI tools, DirectQuery and aggregations.

QUESTION 179

**What are Genie spaces in Databricks?**

A. A CI/CD tool

**B. ✓ A way for business users to ask natural-language questions over curated, governed tables**

C. A cluster type

D. A storage tier

> **Answer: B** Genie turns questions into SQL over the tables and instructions a data team curates, and respects Unity Catalog permissions.

QUESTION 180

**What does AI Gateway add to model serving endpoints?**

A. Vector indexes

B. Faster GPUs

C. Model training

**D. ✓ Central governance: rate limits, guardrails, usage tracking and payload logging**

> **Answer: D** AI Gateway governs access to foundation, external and custom model endpoints.

QUESTION 181

**How can analysts run batch LLM inference (for example summarizing reviews) directly in SQL?**

A. It isn't possible

B. By exporting to Excel

C. With COPY INTO

**D. ✓ With the ai_query() function**

> **Answer: D** ai_query() calls a model serving endpoint from SQL, so you can enrich tables at scale.

## Sizing, Performance & Cost

The DBU pricing model, choosing and sizing compute, SQL warehouse scaling, compute policies and cost monitoring.

QUESTION 182

**For a classic job cluster, which costs appear on the bill?**

A. A flat monthly fee

**B. ✓ Databricks DBUs, plus Azure VM, disk and network costs billed by Azure**

C. Only Azure VMs

D. Only DBUs

> **Answer: B** Classic compute is billed twice: DBUs by Databricks, and infrastructure by Azure. Serverless DBU rates include the infrastructure.

QUESTION 183

**Users report BI dashboards slow down only at 9 a.m., when 200 people open them. Each query on its own is fast. What should you change?**

A. Turn off Photon

**B. ✓ Increase the warehouse's max clusters (scale out)**

C. Increase the warehouse size

D. Use an all-purpose cluster

**Answer: B** Queuing under concurrency is solved by scaling out. Increasing the size helps individual heavy queries.

QUESTION 184

**Where should spot VMs be used in a Databricks cluster?**

**A. ✓ For workers of fault-tolerant batch jobs, with fallback to on-demand**

B. Only for SQL warehouses

C. For the driver of production streaming jobs

D. For every node of every cluster

**Answer: A** Spark can recompute work lost when a worker is evicted. Losing the driver kills the job, so keep it on-demand.

QUESTION 185

**Interactive clusters are left running overnight and there are no limits on node types. What's the best control?**

A. Deleting clusters weekly

B. Email reminders

**C. ✓ Compute policies that enforce auto-termination, max workers, allowed VM types and tags**

D. Removing everyone's workspace access

**Answer: C** Policies make the cost-safe configuration the default and enforce tags so spend can be attributed.

QUESTION 186

**Where do you analyze Databricks spend by tag and SKU with SQL?**

A. information_schema.tables

B. DESCRIBE HISTORY

**C. ✓ system.billing.usage joined with system.billing.list_prices**

D. The Spark UI

**Answer: C** The billing system tables record usage by SKU, workspace and tags. Joining list prices gives you cost.

QUESTION 187

**Photon has a higher DBU rate. When does enabling it reduce cost?**

A. Never

B. Only for Python UDF workloads

**C. ✓ When the speed-up makes total DBU-hours drop by more than the rate increase, which is common for SQL and DataFrame workloads**

D. Only on single-node clusters

> **Answer: C** Compare cost per run. A large speed-up usually more than offsets the higher rate, though UDF-heavy code gains little.

QUESTION 188

**What's included in the DBU price of serverless compute?**

A. Storage costs

B. Networking egress to other clouds

**C. ✓ The compute infrastructure as well, so there's no separate VM bill**

D. Only the Databricks software

> **Answer: C** Serverless DBU rates bundle the infrastructure. Classic compute is billed as DBUs plus Azure VMs.

QUESTION 189

**Idle instances waiting in a cluster pool cost what?**

A. Double DBUs

B. Nothing

**C. ✓ Azure VM costs, but no DBUs**

D. DBUs only

> **Answer: C** Pools keep VMs running so clusters start faster. You pay Azure for those VMs while they're idle.

QUESTION 190

**Which Azure VM family usually suits memory-bound, shuffle-heavy Spark jobs?**

**A. ✓ Memory-optimized (E-series)**

B. Burstable (B-series)

C. GPU (NC-series)

D. Storage-optimized only

> **Answer: A** E-series VMs have more memory per core, which reduces spill in large joins and aggregations.

QUESTION 191

**What do budgets in the Databricks account console do?**

A. Resize clusters

B. Discount DBUs

**C. ✓ Track spend against a threshold and send alerts**

D. Hard-stop all compute at a limit

> **Answer: C** Budgets alert on spend, optionally filtered by workspace or tags. Use policies to enforce limits.

QUESTION 192

**Where do cluster and job custom tags show up?**

**A. ✓ In system.billing.usage, and on the Azure resources for classic compute (so in Azure Cost Management too)**

B. Nowhere

C. Only in audit logs

D. Only in the Spark UI

**Answer: A** Tags flow into billing data, so you can attribute cost by team or project.

QUESTION 193

**What does predictive optimization do for Unity Catalog managed tables?**

A. Lowers DBU prices

B. Predicts query results

**C. ✓ Automatically runs maintenance such as OPTIMIZE and VACUUM when it's worthwhile**

D. Caches every table in memory

**Answer: C** It removes the need for manual maintenance schedules and keeps storage and query cost down.

## Migration, DR & Operations

Migrating warehouses and Hive metastores, disaster recovery, infrastructure as code and CI/CD across environments.

QUESTION 194

**Which Databricks Labs tool helps upgrade a workspace's Hive metastore tables and permissions to Unity Catalog?**

A. Delta Sharing

B. MLflow

**C. ✓ UCX**

D. dbx

**Answer: C** UCX assesses the workspace, migrates tables (for example with SYNC for external tables) and maps permissions to Unity Catalog.

QUESTION 195

**Why is SHALLOW CLONE unsuitable for disaster recovery to another region?**

**A. ✓ It references the source table's data files, so it's useless if the primary storage is lost**

B. It's slower than deep clone

C. It doesn't copy the schema

D. It can't be queried

**Answer: A** A shallow clone copies only metadata. Deep clone copies the data files, and incrementally on later runs.

QUESTION 196

**Why is ADLS geo-redundant storage alone risky for Delta tables during a regional failover?**

A. Delta can't read GRS accounts

**B. ✓ Replication is asynchronous and not ordered per table, so a table's log and data files can be inconsistent after failover**

C. GRS doesn't copy Parquet files

D. GRS deletes old versions

**Answer: B** Delta needs its log and data files to be consistent. Delta-aware copies (deep clone) give a consistent table in the secondary region.

QUESTION 197

**In an active-passive DR design with deep clones every 4 hours, what's the approximate RPO?**

**A. ✓ Up to 4 hours of data**

B. It depends only on the cluster size

C. Zero

D. 4 days

**Answer: A** Anything written since the last successful copy can be lost, so the copy interval roughly sets the RPO.

QUESTION 198

**Which combination is recommended for infrastructure versus jobs and pipelines as code?**

**A. ✓ Terraform for workspaces, networking and Unity Catalog; Databricks Asset Bundles for jobs, pipelines and code**

B. Manual setup in the UI, documented in a wiki

C. Asset Bundles for VNets

D. ARM templates for everything

**Answer: A** Terraform (with the databricks and azurerm providers) manages platform resources. Bundles package and deploy workloads per environment.

QUESTION 199

**What is the most important principle when migrating a large legacy warehouse?**

A. Big-bang cut-over in one weekend

B. Rewrite every report first

**C. ✓ Migrate incrementally by domain, run old and new in parallel, and reconcile before switching consumers**

D. Copy data without validating it

**Answer: C** Incremental, validated migration limits risk and lets the business switch over one domain at a time.

QUESTION 200

**What is the `SYNC` command used for?**

A. Replicating data across regions

B. Synchronizing cluster clocks

C. Syncing Git folders

**D. ✓ Upgrading external Hive metastore tables to Unity Catalog**

**Answer: D** SYNC creates or updates UC external tables from Hive metastore tables without copying data.

QUESTION 201

**You re-run `CREATE OR REPLACE TABLE dr.t DEEP CLONE prod.t`. What does the second run copy?**

A. Only metadata

B. Nothing

**C. ✓ Only files that changed since the last clone (it's incremental)**

D. Everything again

**Answer: C** Deep clone is incremental on re-runs, which makes it practical for scheduled DR copies.

QUESTION 202

**Where should production job and pipeline definitions live?**

A. In cluster init scripts

B. Only in the workspace UI

C. In a spreadsheet

**D. ✓ In Git, as Databricks Asset Bundles deployed by CI/CD**

**Answer: D** Bundles make jobs versioned, reviewable and repeatable across environments.

QUESTION 203

**Who should deploy changes to the production workspace?**

A. Databricks support

B. The workspace admin, by hand

**C. ✓ A CI/CD pipeline authenticating as a service principal**

D. Any developer, from their laptop

**Answer: C** Automated, reviewed deployments with a non-human identity are repeatable and auditable.

QUESTION 204

**What does RTO (recovery time objective) measure?**

A. How much data can be lost

**B. ✓ How long it may take to restore service after a disaster**

C. Query latency

D. Storage replication speed

**Answer: B** RTO is downtime tolerance. RPO is data-loss tolerance.

QUESTION 205

**During a migration, how can you compare source and target data without copying the source?**

**A. ✓ Lakehouse Federation, to query the source database in place**

B. Auto Loader

C. VACUUM

D. Delta Sharing

**Answer: A** Federated queries let you reconcile counts and aggregates against the legacy system directly.

# Answer key

Question number and correct option, in the order they appear above.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1** D | **2** A | **3** A | **4** C | **5** C | **6** A | **7** C | **8** B | **9** B | **10** C |
| **11** D | **12** C | **13** C | **14** A | **15** A | **16** B | **17** B | **18** B | **19** D | **20** A |
| **21** B | **22** C | **23** C | **24** D | **25** A | **26** B | **27** C | **28** D | **29** D | **30** D |
| **31** A | **32** B | **33** D | **34** D | **35** A | **36** B | **37** A | **38** C | **39** D | **40** C |
| **41** A | **42** A | **43** B | **44** A | **45** C | **46** D | **47** C | **48** B | **49** A | **50** B |
| **51** A | **52** D | **53** D | **54** D | **55** C | **56** B | **57** D | **58** D | **59** B | **60** A |
| **61** D | **62** D | **63** B | **64** B | **65** B | **66** D | **67** A | **68** B | **69** B | **70** A |
| **71** D | **72** B | **73** B | **74** C | **75** A | **76** A | **77** B | **78** B | **79** A | **80** D |
| **81** B | **82** A | **83** B | **84** A | **85** B | **86** A | **87** D | **88** D | **89** B | **90** A |
| **91** B | **92** C | **93** A | **94** D | **95** B | **96** B | **97** C | **98** B | **99** B | **100** C |
| **101** D | **102** C | **103** A | **104** D | **105** B | **106** C | **107** A | **108** D | **109** D | **110** A |
| **111** B | **112** D | **113** D | **114** C | **115** D | **116** C | **117** A | **118** C | **119** B | **120** C |
| **121** C | **122** C | **123** B | **124** C | **125** A | **126** D | **127** D | **128** D | **129** A | **130** D |
| **131** A | **132** C | **133** A | **134** A | **135** D | **136** A | **137** B | **138** C | **139** A | **140** B |
| **141** C | **142** A | **143** C | **144** A | **145** D | **146** A | **147** B | **148** D | **149** D | **150** D |
| **151** B | **152** A | **153** B | **154** C | **155** A | **156** B | **157** C | **158** C | **159** B | **160** D |
| **161** A | **162** B | **163** A | **164** D | **165** B | **166** C | **167** B | **168** B | **169** B | **170** A |
| **171** B | **172** B | **173** D | **174** D | **175** C | **176** C | **177** C | **178** B | **179** B | **180** D |
| **181** D | **182** B | **183** B | **184** A | **185** C | **186** C | **187** C | **188** C | **189** C | **190** A |
| **191** C | **192** A | **193** C | **194** C | **195** A | **196** B | **197** A | **198** A | **199** C | **200** D |
| **201** C | **202** D | **203** C | **204** B | **205** A | | | | | |